Spencer Alan Hill

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EDUCATION

2016	Ph.D.	Atmospheric and Oceanic Sciences	Princeton University
2011	B.S.	Atmospheric and Oceanic Sciences; Applied Mathematics	UCLA

PROFESSIONAL APPOINTMENTS

2024- Affiliated Faculty, City University of New York (CUNY) Remote Sensing Earth Systems Institute (CREST)
2024- Doctoral Faculty, Dept. of Physics, CUNY Graduate Center
2023- Doctoral Faculty, Dept. of Earth and Environmental Sciences, CUNY Graduate Center
2023- Affiliated Faculty, Program in Earth System Science and Environmental Engineering, City College of New York (CCNY)
2023- Assistant Professor, Dept. of Earth and Atmospheric Sciences, CCNY
2021-26 Adjunct Associate Research Scientist, Lamont-Doherty Earth Observatory, Columbia University
2021-23 Associate Research Scholar, Program in Atmospheric and Oceanic Sciences, Princeton University
2019-21 Postdoctoral Research Scientist, Lamont-Doherty Earth Observatory, Columbia University
2016-19 Postdoctoral Research Scientist, dual appointment, Division of Geological and Planetary Sciences, California Institute of Technology, and Dept. of Atmospheric and Oceanic Sciences, UCLA
2011-16 Graduate Research Assistant, Program in Atmospheric and Oceanic Sciences, Princeton University

EXTRAMURAL GRANTS AWARDED

2026 "CAREER: Advancing understanding of Hadley cell dynamics from foundational theory to large language models." Principal Investigator. 5 yr, $1,050,883. NSF CAREER program, award 2544433.
2025 "Collaborative Research: Extreme Rainfall in South Asia–Mechanisms of Variability." Principal Investigator. 3 yr, $408,862. NSF Climate and Large-Scale Dynamics, award AGS-2437879.
2024 "The Climate Lighthouse: DOE-CCNY Urban Climate Hazard Resilience Center." Co-Investigator. 3 yr, $998,973. U.S. Dept. of Energy, Office of Biological & Environmental Research, Climate Resilience Centers program.
2023 "How do energy fluxes link precipitation variability across the tropical weather-climate continuum?" Principal Investigator. 2 yr, $399,465. NSF Climate and Large-Scale Dynamics, award AGS-2411723.
[Transfer to CCNY from Princeton of original AGS-2123327 awarded 2021 (3 yr, $682,038)]

CUNY INTERNAL GRANTS AWARDED

2026 "AI-Annotated Research Papers for Climate Literacy." Principal Investigator. 1 yr, $12,000. CUNY OAA AI Innovation Fund.
2025 "Machine learning applied to open problems in Earth and Atmospheric Sciences." Principal Investigator. 1 yr, $10,000. CUNY Research in the Classroom program.
CUNY Critical A.I. Literacy Institute (CALI). 1 yr, $8,000.
"A first-ever quantification of moist static energy transports by tropical cyclones." Principal Investigator. 1 yr, $5,999.53. Professional Staff Congress (PSC) of CUNY Research Award, Cycle 56.

POSTDOCTORAL AND GRADUATE FELLOWSHIPS AWARDED

2019 Columbia University Earth Institute Postdoctoral Research Fellowship (2019-21)
2016 California Institute of Technology Foster and Coco Stanback Postdoctoral Research Fellowship (deferred to 2018-19)
NSF Atmospheric and Geospace Sciences Postdoctoral Research Fellowship (2016-18)
2013 Dept. of Defense National Defense Science and Engineering Graduate Research Fellowship (2013-16)

PUBLICATIONS

Articles submitted/under review/in revision

García-Franco JL, R Chadwick, **SA Hill**, K Inoue, K Schwarzwald, A Varuolo-Clarke. "What drives the Mesoamerican Midsummer Drought? Insights from the moist static energy budget." Submitted to *Climate Dynamics*.

Parsells, AS, **SA Hill**, M Biasutti. "Evaluating the Energetic Framework of Seasonal ITCZ Shifts Across Reanalyses and Computational Methods." Revised for *J. Climate*.

Refereed journal articles

2026 29. Randazzo GS, **SA Hill**, JF Booth, RD Field. "A Wall of Smoke: Understanding the Drivers of the June 2023 Canadian Wildfire Smoke Event for New York City." *J. Geophys. Res. - Atmos.*, **131**(9), e2025JD044798. doi:10.1029/2025JD044798.

2025 28. Zhang, P, NJ Lutsko, **SA Hill**, S.-P. Xie. "Hadley Cell Dynamics in an Axisymmetric Single-Layer Model: Effects of Parameterized Eddies and Equatorial Heating." *J. Atmos. Sci.*, **82**, 2757-2770. doi:10.1175/JAS-D-25-0076.1

27. **Hill, SA**, D Zamir-Meyers, AH Sobel, M Biasutti, MA Cane, MK Tippett, F Ahmed. "More extreme Indian monsoon rainfall in El Niño summers." *Science*, **389**, 1220-1224. doi:10.1126/science.adg5577. Link for free article access.
Press: CCNY press release. Lamont-Doherty press release. Article in *Deccan Herald*.

26. **Hill, SA**, S Bordoni, JL Mitchell, JM Lora. "Interpreting seasonal and interannual Hadley cell descending edge migrations via the cell-mean Rossby number." *J. Climate*, **38**, 5505-5520. doi:10.1175/JCLI-D-24-0678.1

2024 25. Mossel, C, **S Hill**, N Samal, J Booth, N Devineni. "Increasing Extreme Hourly Precipitation Risk for New York City after Hurricane Ida." *Scientific Reports*, **14**, 27947. doi:10.1038/s41598-024-78704-9.

24. Byrne MP, G Hegerl, J. Scheff, …, **SA Hill**, et al. "Theory and the future of land-climate science." *Nature Geoscience*, **17**, 1079-1086. doi:10.1038/s41561-024-01553-8.

23. Clark, JP, P Lin, **SA Hill**. "ITCZ Response to Disabling Parameterized Convection in Global Fixed-SST Aquaplanet Simulations at 50 km and 6 km Resolutions." *J. Adv. in Mod. Earth Sys.*, **16**, e2023MS003968. doi:10.1029/2023MS003968.

2023 22. Zurita-Gotor P, IM Held, TM Merlis, CY Chang, **SA Hill**, CG MacDonald. "Non-uniqueness in ITCZ latitude due to radiation-circulation coupling in an idealized GCM." *J. Adv. in Mod. Earth Sys.*, **15**(10), e2023MS003736. doi: 10.1029/2023MS003736.

21. Biasutti M, M Ting, **SA Hill**. "The dynamics and changes of the world's monsoons." *Physics Today*, **76**(9), 32-38. doi: 10.1063/PT.3.5308.

20. Moscoso JE, RE Tripoli, S Chen, H Gonzalez, **SA Hill**, N Khoo, TL Lonner, JM Aurnou. "Low-cost table-top experiments for teaching multi-scale geophysical fluid dynamics." *Frontiers in Marine Science*, **10**, 1192056. doi: 10.3389/fmars.2023.1192056.

19. Ahmed F, JD Neelin, **SA Hill**, K Schiro, and H Su. "A process model for ITCZ narrowing under warming highlights clear-sky water vapor feedbacks and gross moist stability changes in AMIP models." *J. Climate*, **36**, 4913-4931. doi: 10.1175/JCLI-D-22-0689.1.

18. Mahfouz N, **SA Hill**, H Guo, and Y Ming. "The radiative and cloud responses to sea salt aerosol geoengineering in GFDL models." *Geophys. Res. Lett.*, **50**, e2022GL102340. doi: 10.1029/2022GL102340.

2022
17. **Hill SA**, NJ Burls, A Fedorov, and TM Merlis. "Symmetric and antisymmetric components of polar-amplified warming." *J. Climate*, **35**, 3157-3172. doi: 10.1175/JCLI-D-20-0972.1.
16. **Hill SA**, S Bordoni, and JL Mitchell. "A theory for the Hadley cell descending and ascending edges throughout the annual cycle." *J. Atmos. Sci.*, **79**, 2515-2528. doi: 10.1175/JAS-D-21-0328.1.
15. Biasutti M, **SA Hill**, and A Voigt. "The Effect Of An Equatorial Continent On The Tropical Rain Belt. Part 2: Summer Monsoons." *J. Climate*, **35**, 3091–3107. doi: 10.1175/JCLI-D-21-0588.1
14. **Hill SA**, AH Sobel, M Biasutti, and MA Cane. "On the all-India rainfall index and sub-India rainfall heterogeneity." *Geophys. Res. Lett.*, **49**, e2021GL096541, 10 pp. doi: 10.1029/2021GL096541.

2021
13. **Hill SA**, S Bordoni, and JL Mitchell. "Solsticial Hadley Cell ascending edge theory from supercriticality." *J. Atmos. Sci*, **78**, 1999-2011. doi: 10.1175/JAS-D-20-0341.1.
12. Mitchell, JL and **SA Hill**. "Constraints from invariant subtropical vertical velocities on the scalings of Hadley cell strength and downdraft width with rotation rate." *J. Atmos. Sci*, **78**, 1445-1463. doi: 10.1175/JAS-D-20-0191.1

2020
11. **Hill SA**, S Bordoni, and JL Mitchell. "Axisymmetric Hadley Cell theory with a fixed tropopause temperature rather than height." *J. Atmos. Sci.*, **77**, 1279-1294. doi: 10.1175/JAS-D-19-0169.1.

2019
10. **Hill SA**. "Theories for past and future monsoon rainfall changes." *Curr. Clim. Change Rep.*, **5**, 160-171. doi: 10.1007%2Fs40641-019-00137-8.
9. **Hill SA**, S Bordoni, and JL Mitchell. "Axisymmetric constraints on cross-equatorial Hadley cell extent." *J. Atmos. Sci.*, **76**, 1547-1564. doi: 10.1175/JAS-D-18-0306.1.

2018
8. **Hill SA**, JM Lora, N Khoo, SP Faulk, and JM Aurnou. "Affordable rotating fluid demonstrations for geoscience education: The *DIYnamics* project." *Bull. Am. Met. Soc.*, **99**, 2529-2538. doi: 10.1175/BAMS-D-17-0215.1.
7. **Hill SA**, Y Ming, and M Zhao. "Robust responses of the Sahelian hydrological cycle to global warming." *J. Climate*, **31**, 9793-9814. doi: 10.1175/JCLI-D-18-0238.1.
6. Smyth J, **SA Hill**, and Y Ming. "Simulated responses of the West African monsoon and zonal-mean tropical precipitation to early Holocene orbital forcing." *Geophys. Res. Lett.*, **45**, 12,049-12,057. doi: 10.1029/2018GL080494.

2017
5. **Hill SA**, Y Ming, IM Held, and M Zhao. "A moist static energy budget-based analysis of the Sahel rainfall response to uniform oceanic warming." *J. Climate*, **30**, 5637-5660. doi: 10.1175/JCLI-D-16-0785.1.
4. Brown PT, Y Ming, W Li, and **SA Hill**. "Change in the magnitude and mechanisms of unforced low-frequency surface temperature variability in a warmer climate." *Nature Climate Change*, **7**, 743-748. 10.1038/nclimate3381.
3. Jeevanjee N, P Hassanzadeh, **SA Hill**, and A Sheshadri. "A perspective on climate model hierarchies." *J. Adv. in Mod. Earth Sys.*, **9**, 1760-1771. doi: 10.1002/2017MS001038.

2015
2. **Hill SA**, Y Ming, and IM Held. "Mechanisms of forced tropical meridional energy flux change." *J. Climate*, **28**, 1725-1742. doi: 10.1175/JCLI-D-14-00165.1.
   Corrigendum: https://dx.doi.org/10.1175/JCLI-D-16-0485.1.

2012
1. **Hill SA** and Y Ming. "Nonlinear climate response to regional brightening of tropical marine stratocumulus." *Geophys. Res. Lett.*, **39**, L15707, 5 pp. doi: 10.1029/2012GL052064.

Invited book chapters

2026 **Hill SA**. "Hadley circulation and trade winds: Mean seasonal cycle and interannual variability." In *Encyclopedia of Climate System Science*, Elsevier. doi:10.1016/B978-0-443-15748-6.00112-X.

Meeting summaries

2026 Byrne MP, GC Hegerl, J Scheff, …, **SA Hill**, et al. "Deepening Mechanistic Understanding of Land Climate." *Bull. Am. Met. Soc.*, early online release. doi:10.1175/BAMS-D-26-0341.1.

Book reviews

2012 **Hill SA** "A head in the clouds elucidates climate" (book review of *Atmosphere, Clouds, and Climate* by David Randall). *Science*, **337**, 1 pp., doi: 10.1126/science.1225615.

Other publications

2022 Biasutti M, M Ting, and **SA Hill**. "How the south Asian monsoon is changing in a warming climate." Guest post, *Carbon Brief*. Sep. 15th. https://www.carbonbrief.org/guest-post-how-the-south-asian-monsoon-is-changing-in-a-warming-climate/.

2017-2022 Fourteen blog posts for the DIYnamics blog. Available at https://diynamics.github.io/blog/author/spencer-hill.html.

AWARDS AND HONORS

2021 Columbia University Nominee, Blavatnik Regional Award for Young Scientists
2012 Princeton University Elliotte Robinson Little '25 Fellowship
2011 American Meteorological Society Annual Meeting Climate Change Travel Scholarship
NSF Graduate Research Fellowship Honorable Mention
UCLA Magna Cum Laude and College Honors graduation distinctions
2009 National Oceanic and Atmospheric Administration Ernest F. Hollings Undergraduate Scholarship
2007 United States Presidential Scholar, conferred by the U.S. Dept. of Education Commission on Presidential Scholars

INVITED COLLOQUIA AND SEMINARS

2026 COASt group seminar, School of Earth and Environmental Sciences, University of St. Andrews, Scotland. Jun. 24th.
2 lectures, Moist Convective Dynamics of Monsoons II workshop, International Center for Theoretical Sciences, Bangalore, India
"The ITCZ energetic framework" May 20th.
"The Hadley circulation" May 21st.
2025 3 lectures, International Center for Theoretical Physics Workshop on Global Monsoons: Theory, Models and Observations, Universidad Nacional Autónoma de México, Mexico City
"Introduction to tropical general circulation and the global monsoons." Jan. 20th.
"Theories for past and future monsoon rainfall changes." Jan. 21st.
"Interannual variability in extreme daily monsoon rainfall." Jan. 22nd.
2024 Dept. of Atmospheric and Oceanic Sciences, University of Wisconsin, Madison, Oct. 21st.
Dept. of Earth and Environmental Sciences, City University of New York Graduate Center, Apr. 4th.
Division of Ocean and Climate Physics seminar series, Lamont-Doherty Earth Observatory, Mar. 29th.
Dept. of Mechanical Engineering, City College of New York, Feb. 22nd.
Dept. of Physics, City College of New York, Feb. 7th.
2023 Dept. of Earth and Planetary Science, Yale University, Mar. 8th.
Atmosphere, Ocean, and Climate Dynamics seminar series, Yale University, Mar. 9th.
Dept. of Atmospheric and Oceanic Sciences, UCLA, Mar. 1st.
Dept. of Earth and Atmospheric Sciences, City College of New York, Feb. 14th.
2022 Equilibrium Climate Sensitivity & Cloud Feedback Virtual Symposium, Mar. 24th.
2021 Atmosphere, Ocean, and Climate Dynamics seminar series, Yale University, Feb. 25th.
Meteorology Seminar Series, Dept. of Earth, Atmospheric, and Oceanic Sciences, Florida State University, Feb. 18th.
2019 Monthly Climate Meeting, Earth Research Institute, University of California – Santa Barbara, Jan. 7th.
2018 Division of Ocean and Climate Physics, Lamont-Doherty Earth Observatory, Feb. 16th.

NOAA Geophysical Fluid Dynamics Laboratory, Feb. 14th.

2016 Center for Atmosphere Ocean Science, New York University, Nov. 9th.

Natural and Behavioral Sciences Lecture, Gaede Institute for the Liberal Arts, Westmont College, Oct. 13th.

2015 Student seminar series, Center for Atmospheric and Oceanic Sciences, New York University, Mar. 6th.

2014 Dept. of Geophysics, Yale University, Oct. 9th.

CONFERENCE ACTIVITIES

Chaired sessions

2026 "Tropical Atmospheric and Oceanic Dynamics" Co-chair. Joint session, AMS Atmospheric and Oceanic Fluid Dynamics Conference and AMS Symposium on Tropical Meteorology and Climate, Houston, TX, Jan. 26th.

2024 "Tropical Atmospheric Dynamics." AMS Atmospheric and Oceanic Fluid Dynamics Conference, Burlington, VT, Jun. 28th.

2023 "Large-Scale Atmospheric Dynamics and Climate: Jet Streams, Storm Tracks, Stationary Waves, and Monsoons IV" AMS Annual Meeting, Denver, CO, Jan. 10th.

2022 "Monsoon Dynamics: Variability, Change, and Impacts I and II" AMS Annual Meeting (virtual), Jan. 26th-27th.

2018 "Monsoons: Observations, Subseasonal, Seasonal, and Interannual to Decadal Variability, Forecast, Climate Change, and Extremes III." AGU Fall Meeting, Washington, D.C., Dec. 11th.

2017 "Idealized approaches to the atmospheric and oceanic circulation II." American Meteorological Society 21st Conference on Atmospheric and Oceanic Fluid Dynamics, Portland, OR, Jun. 26th.

2016 "Tropical circulations and their sensitivities to changes in climate I." AGU Fall Meeting, San Francisco, CA, Dec. 16th.

"Tropical convection and radiative convective equilibrium." World Climate Research Programme Model Hierarchies Workshop, Princeton, NJ, Nov. 3rd.

Invited Conference Talks

2023 "More extreme Indian monsoon daily rainfall in El Niño summers." AMS Annual Meeting, Denver, CO, Jan. 9th.

2022 "Indian summer monsoon rainfall interannual variability: seasonal-mean and daily extremes." Continental Climate Change Workshop, St. Andrews, Scotland, Jun. 8th.

2019 "Toward an analytical, predictive theory for the location of Hadley and monsoonal cell ascending branches." AGU Fall Meeting, San Francisco, CA, Dec. 10th.

2016 "`infinite-diff` and `animal-spharm`: `xarray`-based finite differencing and spherical harmonics." Columbia University Python and Atmospheric and Oceanic Sciences Workshop, New York, NY, Nov. 12th.

2015 "Towards constraining Sahel rainfall responses to global mean temperature changes." Linde Center for Global Environmental Science, California Institute of Technology, Monsoons: Past, Present and Future workshop, Pasadena, CA, May 21st.

Other Conference Presentations

2026 "A generalized equal-area model." Talk. AGU Annual Meeting. San Francisco, CA. Dec. 8th.

"ENSO influences on monsoon low-pressure system circulation and extreme rainfall over Indian land." Land-Climate Workshop, St. Andrews, Scotland, Jun. 18th.

"A generalized equal-area model." AMS Atmospheric and Oceanic Fluid Dynamics. Houston, TX. Jan. 26th.

2025 "More extreme Indian monsoon daily rainfall in El Niño summers." Talk. Eighth World Meteorological Organization Workshop on Monsoons. Virtual. Mar. 18th.

2024 "Explaining Hadley cell extent across timescales via the cell-mean Rossby number." Talk. AGU Annual Meeting. Washington, DC. Dec. 13th.

"More extreme Indian monsoon daily rainfall in El Niño summers." Talk. 4th Workshop on Convective Organization. International Centre for Theoretical Physics, Trieste, Italy. Jul. 11th.

"Investigating Hadley cell extent across timescales via the cell-mean Rossby number." Talk. AMS Atmospheric and Oceanic Fluid Dynamics meeting. Burlington, VT. Jun. 28th.

"More extreme Indian monsoon daily rainfall in El Niño summers." Poster. CFMIP Meeting. Boston, MA. Jun. 4th.

"The Hadley Cells Across Seasons and the Solar System" Talk. AMS Annual Meeting. Baltimore, MD. Jun. 17th.

2023 "More extreme Indian monsoon daily rainfall during El Niño summers." Poster. AGU Fall Meeting. San Francisco, CA. Dec. 12th

2022 "Hadley cell extent across seasons and the solar system." Talk. AMS Atmospheric and Oceanic Fluid Dynamics Meeting. Breckenridge, CO. Jun. 17th.

"Symmetric and antisymmetric components of polar-amplified warming." Poster. U.S. CLIVAR Pattern Effect Workshop, Boulder, CO. May 11th.

"On the All-India Rainfall Index, Sub-India Rainfall Heterogeneity, ENSO, and Teleconnections from the Indian Ocean" Talk. AMS Annual Meeting (virtual). Jan. 26th.

"Symmetric and Antisymmetric Components of Polar-Amplified Warming." Talk. AMS Annual Meeting (virtual). Jan. 26th.

2021 "New scalings for the ascending and descending branch positions of the solsticial Hadley cells in planetary atmospheres including Titan" Poster. AGU Fall Meeting. New Orleans, LA. Dec. 16th.

"On the all-India rainfall index, sub-India rainfall heterogeneity, ENSO, and teleconnections from the Indian Ocean" Talk. AGU Fall Meeting. New Orleans, LA. Dec. 15th.

"Connecting sub-India, sub-seasonal monsoon rainfall variability with all-India, all-summer monsoon rainfall." AMS Annual Meeting (virtual). Talk. Jan. 14th.

2020 "Sub-India summer monsoon rainfall variability and its implications for all-India summer monsoon rainfall prediction." AGU Fall Meeting (virtual). Poster. Dec. 1st-17th.

2019 "Simulated polar amplification and its causes on decadal to millennial timescales." Poster. AGU Fall Meeting. San Francisco, CA. Dec. 10th.

"Modernizing Axisymmetric Hadley Cell and Monsoon Theory." Talk. AMS 22nd Conference on Atmospheric and Oceanic Fluid Dynamics. Portland, ME. Jun. 25th.

2018 "What sets the locations of the solsticial cross-equatorial Hadley cell edges?" Talk. AGU Fall Meeting. Washington, DC. Dec. 13th.

"Towards transient simulation of the Green Sahara onset and demise through idealized modeling of vegetation-land-atmosphere interactions." Poster. 17th Swiss Climate Summer School: Earth system variability through time. Grindelwald, Switzerland. Aug. 28th.

"What Determines the ITCZ Position During Solsticial Seasons on Earth and Other Planets?" Talk. AMS 33rd Conference on Hurricanes and Tropical Meteorology. Ponte Vedra, FL. Apr. 16th.

2017 "Dry Rainbelts: Understanding Boundary Layer Controls on the ITCZ Using a Dry Dynamical Core." Talk. AGU Fall Meeting. New Orleans, LA. Dec. 14th.

"Towards transient simulation of the Green Sahara onset and demise through idealized modeling of vegetation-land-atmosphere interactions." Poster. Gordon Research Conference on Radiation and Climate. Bates College, Lewiston, ME. Jul. 19th.

"Control of convergence zone migrations by planetary parameters." Poster. AMS 21st Conference on Atmospheric and Oceanic Fluid Dynamics. Portland, OR. Jun. 27th.

"Automate your climate and weather data analysis with aospy." Talk. AMS Annual Meeting, Seattle, WA. Jan. 24th.

"Energetic and precipitation responses in the Sahel to sea surface temperature perturbations." Talk. AMS Annual Meeting, Seattle, WA. Jan. 24th.

2016 "Robust drying influence of mean ocean surface warming on The Sahel and implications for constraining future rainfall change." Poster. AGU Fall Meeting, San Francisco, CA. Dec. 16th.

"A hierarchy of perturbation complexites: Case study of Sahel rainfall response to global warming" Poster. WCRP Model Hierarchies Workshop, Princeton University, Princeton, NJ. Nov. 2nd.

2015 "Towards constraining future rainfall in the Sahel using the moist static energy budget." Talk. AGU Fall Meeting, San Francisco, CA. Dec. 14th.
"Convection scheme, cloud, and stability effects on Sahel rainfall response to uniform warming." Poster. AMS Annual Meeting, Phoenix, AZ. Jan. 6th.
2014 "Convection scheme, cloud, and stability effects on Sahel rainfall response to uniform warming." Poster. AGU Fall Meeting, San Francisco, CA. Dec. 15th.
"Mechanisms of forced tropical meridional energy flux change." Poster. Latsis Symposium, ETH Zurich, Zurich, Switzerland. Jun. 19th.
"Mean and extreme tropical precipitation changes caused by the uniform and spatially varying components of anthropogenic forcing." Talk. AMS Annual Meeting, Atlanta, GA. Feb. 5th.
2013 "Mechanisms of forced tropical meridional energy flux change." Talk. AGU Fall Meeting, San Francisco, CA. Dec. 13th.
"Mechanisms of forced tropical meridional energy flux change." Talk. Graduate Climate Conference, Woods Hole Oceanographic Institution, Woods Hole, MA. Nov. 2nd.
"Mechanisms of forced tropical meridional energy flux change." Poster presentation. Gordon Research Conference, Colby-Sawyer College, New London, NH. Jul. 9th.
2012 "Climate response to a geoengineered brightening of subtropical marine boundary clouds." Poster. 11th Annual Student Conference at the AMS Annual Meeting, New Orleans, LA. Jan. 22nd.
2010 "Climate response to a geoengineered brightening of subtropical marine boundary clouds." Poster. San Francisco, CA. Dec. 14th.
"Climate response to a geoengineered brightening of subtropical marine boundary clouds." Talk. Special Symposium on Aerosols in Geoengineering at the American Association for Aerosol Research 29th Annual Conference. Portland, OR. Oct. 26th.
"Investigating climate response to geoengineering using a global climate model." Talk. National Oceanic and Atmospheric Administration Office of Education Science Symposium, Silver Spring, MD. Aug. 3rd.

CAMPUS AND DEPARTMENTAL TALKS

2026 "More extreme Indian monsoon rainfall during El Niño summers." CCNY Research & Creative Works Day. Nov. 9th.
Princeton AOS Dynamics Group Seminar. Feb. 24th.
2025 CCNY Harlem 2032 Faculty Brown-Bag seminar series. Oct. 30th.
CCNY Mini Workshop on Convection. Oct. 22nd.
2023 Princeton AOS Dynamics Group Seminar. Feb. 23rd.
2022 Princeton AOS Dynamics Group Seminar. Aug. 11th.
Princeton AOS Discussion group on Isaac Held's Blog. Aug. 1st.
Princeton AOS Tutorial Series for summer interns. Jun. 21st.
2019 Lamont-Doherty Earth Observatory Postdoc Symposium (poster). Sep. 11th.
2018 Dept. of Atmospheric and Oceanic Sciences, UCLA. Nov. 7th.
2016 Division of Geological and Planetary Sciences, California Institute of Technology. Oct. 26th.
Dept. of Atmospheric and Oceanic Sciences, UCLA. Oct. 5th.
2015 Dynamics Seminar Series, Program in Atmospheric and Oceanic Sciences, Princeton University. Mar. 13th.
2012 Student/Postdoc Seminar Series, Princeton AOS. Feb. 28th.
2011 Graduate Research Symposium, Dept. of Geosciences, Princeton University. Nov. 11th.

ADVISING

Postdocs

2025- Shreya Keshri, Dept. of Earth and Atmospheric Sciences, CCNY
2024-26 Haochang Luo, Dept. of Earth and Atmospheric Sciences, CCNY

PhD. students

2025- Gregory Randazzo, Dept. of Earth and Environmental Sciences, CUNY Graduate Center (co-advised with James Booth)
2022- Alexander Parsells, Dept. of Earth and Environmental Sciences, Columbia University (co-advised with Michela Biasutti)

Masters students

2025- Edda Monique Hobuss, Dept. of Earth and Atmospheric Sciences, CCNY
2024- Michelle Wagner, Dept. of Earth and Atmospheric Sciences, CCNY
2023-25 Gregory Randazzo, Dept. of Earth and Atmospheric Sciences, CCNY (co-advised with James Booth)

Undergraduate students

2024-25 Edda Hobuss, CCNY
2021 Destiny Zamir Meyers, Columbia University
Matthew Donahue, Columbia University
2020 Valentina Rojas-Posada, Barnard University (partial funding awarded by competition from Columbia University Earth Institute)
2017 Norris Khoo, UCLA
Micah Kim, UCLA
2016 Juliet Olsen, UCLA

Undergraduate summer interns, as co-advisor

2024 Rachel Ioffe (CCNY IT-ROCS REU program, co-advisor Greg Randazzo)
Ilvis Cid (CCNY IT-ROCS REU program, co-advisor Greg Randazzo)
2023 Kittson Hamill (NOAA GFDL, co-advisor Nadir Jeevanjee)
2022 Rea Restugi (NOAA GFDL, co-advisors Akshaya Nikumbh, Yi Ming)
2015 Jane Smyth (NOAA GFDL, co-advisor Yi Ming)
2014 Marjahn Finlayson (NOAA GFDL, co-advisor Yi Ming)
2013 Colin Raymond (NOAA GFDL, co-advisor Yi Ming)

TEACHING ACTIVITIES

Courses

2026 Fall (co-taught) CCNY EAS 31182 “Music and the Problem of Climate: Intersections of Science, Sound, and Culture.”
2026 Fall (co-taught) CUNY EES 71600 “Earth System Science I.”
2026 Spring CCNY EAS 48800/B8800 “Climate and Climate Change”
2025 Fall CCNY EAS 42000/A4200 “Statistical Methods / Quantitative Data Analysis in Earth and Atmospheric Sciences.”
2025 Fall (co-taught) CUNY EES 71600 “Earth System Science I.”
2025 Spring CCNY EAS 48800/B8800 “Climate and Climate Change”
2024 Fall CCNY EAS 30000/B1300 “Earth and Environmental Science Seminar”
2024 Spring CCNY EAS 48800/B8800 “Climate and Climate Change”
2023 Fall CCNY EAS 42000/A4200 “Statistical Methods / Quantitative Data Analysis in Earth and Atmospheric Sciences.”

Teacher training workshops organized

2022 "Teaching atmosphere, ocean, and planetary fluid dynamic fundamentals vividly with rotating tanks." Earth Educators Rendezvous, Twin Cities, MN, Jul. 14-15th.

Teaching Assistant

2014 "Physics of Earth: The Habitable Planet." Upper-division undergraduate course. Dept. of Geosciences, Princeton University.

Certifications

2016 Teaching Transcript Certification, McGraw Center for Teaching and Learning, Princeton University.

Guest Lectures/Labs

2026 Lecture for "Climate Change and Society" (EESC 1147) undergraduate course, Boston College. Nov. 20th.
2024 Lecture for "Selected Topics in Advanced Physics: Fluid Dynamics Applied to Classical & Quantum Systems" (PHYS 85200) graduate course, Dept. of Physics, CUNY Graduate Center. Nov. 5th.
Two lectures for "ESS Modeling/Databases" (EAS 30800) undergraduate course, Dept. of Earth and Atmospheric Sciences, City College of New York. Sep. 3rd and 5th.
2023 Lecture for "Climate Thermodynamics and Energy Transfer" (EESC 4040) graduate course, Dept. of Earth & Environmental Science, Columbia University. Apr. 25th.
2022 Lab for EAS 30900/B3090 "Fundamentals of Atmospheric Science" graduate course, Dept. of Earth and Atmospheric Sciences, City College of New York. Dec. 5th.
Lab for "Earth's Environmental Systems: The Climate System" undergraduate course, Dept. of Earth and Environmental Sciences, Columbia University. Feb. 24th.
2018 Three lectures for "Introduction to Atmospheric and Oceanic Fluids," graduate course, Dept. of Atmospheric and Oceanic Sciences, UCLA.
"The general circulation of the atmosphere: Energetics." Nov. 13th.
"The general circulation of the atmosphere: Momentum." Nov. 15th.
"The general circulation of the atmosphere: Lab." Nov. 29th.
2014 Two lectures for "Physics of Earth: The Habitable Planet," upper-division undergraduate course, Dept. of Geosciences, Princeton University.
"Weather: From orbital order, atmospheric chaos." Oct. 14th.
"Climate modeling and climate change projections." Dec. 10th.

SCIENCE OUTREACH AND MEDIA

2026 "How El Niño Affects Weather from Inwood to India." Public talk, Taste of Science NYC event "Under Pressure: Decoding the Weather Machine," Ryan's Daughter, New York, NY. Oct. 21st.
"A Wall of Smoke: Understanding the Drivers of the June 2023 Canadian Wildfire Smoke Event for New York City." Virtual talk, Weather and Climate Livestream event. Jun. 3rd.
https://www.youtube.com/live/TzOi_kvajnc?si=SbRiADamrcK3yNhd
2025 "Weather on a table top (why it matters that Earth spins)." Virtual talk, Weather and Climate Science Livestream event. May 31st. https://www.youtube.com/live/RlXBjfIdl-I
2023 "Climate, Chaos and CCNY: A new professor in EAS — Dr. Spencer Hill." Interview, *The RICC* (Research and Innovation at City College) magazine. Oct. 24th.
https://thericc.com/climate-chaos-and-ccny-a-new-professor-in-eas-dr-spencer-hill/
Live on-air radio interview, KCBS San Francisco. Jul. 15th.
2022 City College of New York Physics Club event with Harlem middle school students. Nov. 18th.
2021 "Ask Me Anything" virtual Q+A session, "AMA Session on Climate Action," Presidential Scholars Alumni Association. Oct. 28th.

Interview in *The Medallion* (Newsletter of the Presidential Scholars Alumni Association). "Scholars in Climate Science: Observations from the Field." Sep. 27th.
U.S. Presidential Scholars Foundation roundtable on human and planetary health, Jul. 21st.
"You Spin Me Right Round." Zoom-based presentation on planetary fluid flows for K-12 students, part of the EI Live K12 series, Earth Institute, Columbia University. Feb. 11th. https://www.earth.columbia.edu/videos/view/you-spin-me-right-round

2023- Member, *DIY*namics (organization advancing rotating tank-based geoscience teaching; https://diynamics.github.io).

2017-23 Co-founder and Co-Director, *DIY*namics

2015- Volunteer at many science outreach events at schools or open to the public

SERVICE ACTIVITIES

Thesis committees

2026 Yuliya Selevich, qualifying paper (second grader), Dept. of Earth and Environmental Sciences, CUNY Graduate Center. Aug..

2025 Albana Hoxha, M.S. thesis defense, Dept. of Earth and Atmospheric Sciences, City College of New York. May 13th.
Yuliya Selevich, M.S. thesis defense, Dept. of Earth and Atmospheric Sciences, City College of New York. May 13th.

Professional

Associate Editor, *Journal of the Atmospheric Sciences*, 2023-.

Review panelist, NSF CAREER program, Nov. 2026; NSF GEO EMBRACE (Directorate for Geosciences EMpowering BRoader Academic Capacity and Education), Mar. 2024.

Proposal referee: NSF Climate and Large-Scale Dynamics; NSF Cyberinfrastructure for Sustained Scientific Innovation; NSF EarthCube; NASA Juno Participating Scientist Program; PSC-CUNY Research Award Program

Article referee, *Nature*, *Nature Climate Change*, *Science Advances*, *Nature Communications*, *npj Climate and Atmospheric Science*, *Journal of Advances in Modeling Earth Systems*, *Journal of Climate*, *Current Climate Change Reports*, *Quarterly Journal of the Royal Meteorological Society*, *Geophysical Research Letters*, *Journal of the Atmospheric Sciences*, *Surveys in Geophysics*, *Paleoceanography and Paleoclimatology*, *Geoscientific Model Development*, *Climatic Change*, *Climate Dynamics*, *Journal of Geophysical Research - Atmospheres*, *Weather and Climate Dynamics*, *International Journal of Climatology*, *Journal of Geoscience Education*, *Atmospheric Science Letters*, *Atmospheric Chemistry and Physics*, *Climate Research*

Advisory Board Member, "Building Capacity for Investigating the Use of Spatial Reasoning in Fluid-Earth Science Disciplines," NSF Building Capacity in STEM Education Research (BCSER) program. PI Peggy McNeal. 2023-2024.

Developer and maintainer, `aospy` open-source Python software package for automating analyses of climate datasets (https://aospy.readthedocs.io)

Code contributor, `xarray` package for labeled multidimensional arrays in Python

Departmental/University

2026- Member, faculty search committee, tenure-track Assistant or Associate Professor in Atmospheric Science (Spergel Initiative cluster hire), CCNY EAS

2026 Presentation to undeclared students, CCNY Student Academic Success Hub ("The Hub") Majors Decoded event. Oct. 15th.
Rotating-tank demonstration with James Booth, CCNY EAS Department Open House. Sep. 17th.

2025- Member, CCNY EAS Curriculum Committee

2025 Presentation to new students, CCNY Division of Science Orientation event. Aug. 20th.
Organizer, CCNY EAS Machine Learning summer working group

2024 Presentation to new students, CCNY Division of Science Orientation event. Aug. 21st.
Organizer, CCNY EAS Machine Learning summer reading group
Presentation to undeclared students, CCNY Academic Awareness ("The Hub") event. Mar. 19th.
Presentation to new students, CCNY Division of Science Orientation event. Jan. 23rd.
2015 Organizer, Convection Journal Club, Program in Atmospheric and Oceanic Sciences, Princeton University
2013-15 Organizer, Climate Sensitivity Journal Club, NOAA Geophysical Fluid Dynamics Laboratory
2012-13 Organizer, Student/Postdoc Seminar Series, Program in Atmospheric and Oceanic Sciences, Princeton University
2012-13 Representative to the faculty, Program in Atmospheric and Oceanic Sciences, Princeton University

ADDITIONAL TRAINING

2025 CUNY Critical AI Literacy Institute workshops. Jun. 11-12th; Aug. 19th.
2024 "Building Swift Meaningful Connections through Structured Conversations." Virtual workshop, Heterodox Academy, Oct. 23rd
Workshop on Sustained Dialogue, CCNY, Aug. 26th
"Harnessing AI to Enhance Constructive Disagreement on Campus." Virtual workshop, Heterodox Academy, Aug. 21st
"AI and Pedagogy." Virtual workshop, Teaching and Learning Center, City College of New York, Apr. 4th
2021 Workshop on Inclusive and Culturally Competent Teaching Strategies in the Earth Sciences, Columbia University, Jan. 8th
2020 Racial Sensitivity Workshop, Columbia University Earth Institute, Jun. 30th
2018 17th Swiss Climate Summer School, "Earth System Variability Through Time," Grindelwald, Switzerland
2012-13 Member, Princeton University Energy and Climate Scholars
2012 NOAA Geophysical Fluid Dynamics Laboratory Summer School on Atmospheric Modeling

PROFESSIONAL MEMBERSHIPS

2010- American Geophysical Union
2011- American Meteorological Society
2021- National Association of Geoscience Teachers
2024- Heterodox Academy